FOUNDATIONS OF PARASITOLOGY

A Comprehensive Reviewer in Parasitology

By BELLAMY

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Comprehensive Parasitology Guide (Expanded)

🧬 I. Extended Definition of Terms (Core Concepts)

Term	Comprehensive Definition	Key Characteristics & Clinical Relevance
Parasitism / Symbiosis	**Parasitism:** One benefits (+), host harmed (-). **Mutualism:** Both benefit (+/+). **Commensalism:** One benefits, other unaffected (+/0).	*Ascaris lumbricoides* (Parasitism); *Entamoeba coli* gut flora (Commensalism).
Endoparasite vs. Ectoparasite	**Endo:** Lives inside host (causes "infection"). **Ecto:** Lives on outside of host (causes "infestation").	*Giardia lamblia* (Endo); *Pediculus humanus* (Ecto).
Obligate vs. Facultative	**Obligate:** Cannot complete life cycle without a host. **Facultative:** Free-living but can become parasitic accidentally.	*Plasmodium falciparum* (Obligate); *Naegleria fowleri* (Facultative).
Accidental vs. Opportunistic	**Accidental:** Infects an unusual host. **Opportunistic:** Harmless to healthy individuals, deadly to immunocompromised.	*Toxocara canis* in humans (Accidental); *Cryptosporidium parvum* in HIV (Opportunistic).
Life Cycle	The complete, continuous phase of growth and development from one generation to the next.	Requires understanding to disrupt transmission and apply correct diagnostics.
Direct/Indirect Cycle	**Direct:** Requires only definitive host. **Indirect:** Requires intermediate host(s).	*Trichuris trichiura* (Direct); *Schistosoma mansoni* (Indirect).
Pre-patent Period	The time between infection and the earliest time the parasite can be recovered from the patient.	Also known as **biological incubation period**. Defines when tests become positive.
Incubation Period	The time interval between the entry of the infectious agent and the onset of clinical signs/symptoms.	Also known as **clinical incubation period**.
Monoecious (Hermaphrodite)	Organism possessing both male and female reproductive organs in a single individual.	Most Trematodes (flukes) and all Cestodes (tapeworms).
Dioecious	Having separate male and female individuals.	All Nematodes (roundworms) and the flukes of genus *Schistosoma*.
Coprozoic / Spurious	A free-living organism ingested and passed through the GI tract without establishing infection.	Plant nematodes found in human stool after eating veggies.
Vector (Bio vs. Mech)	Arthropods that transmit parasites. **Biological:** Parasite develops/multiplies inside vector. **Mechanical:** Passive transport.	*Anopheles minimus* (Bio); *Musca domestica* carrying cysts (Mech).

🧍 II. Host Types & Specifics

Concept	Definition & Role	Examples
Definitive Host	Host where parasite reaches sexual maturity and reproduces sexually.	Humans (for *Taenia solium*), Mosquito (for *Plasmodium falciparum*).
Intermediate Host	Harbors larval/asexual stage. Essential for life cycle completion.	Snails (for Flukes), Humans (for *Plasmodium vivax*).
Paratenic / Transport Host	Parasite survives but does not develop. Serves as a vehicle to reach definitive host.	Boar carrying *Ascaris lumbricoides* eggs passively.
Reservoir Host	Animal host that harbors the same parasite species as humans; acts as infection source.	Pigs (*Balantidium coli*), Rodents (*Leishmania donovani*).
Oviparous / Viviparous	**Oviparous:** Lays eggs. **Ovoviviparous:** Eggs hatch inside, releases larvae. **Viviparous:** Live birth.	*Ascaris lumbricoides* (Ovi), *Strongyloides stercoralis* (Ovovivi), *Wuchereria bancrofti* (Vivi).
Autoinfection	Host is repeatedly infected by the parasite already inside them. Causes **hyperinfection**.	*Strongyloides stercoralis*, *Capillaria philippinensis*, *Enterobius vermicularis*.

💩 III. Specimen Collection & Transport

Guideline	Details & Rationale
General Rules	Clean, dry, wide-mouth container. No urine (destroys trophozoites), water (free-living amoeba), or soil (environmental helminths).
Amount & Timing	Solid: 2-5g (walnut size). Liquid: 5-10 mL. Usually 3 samples on alternate days to increase detection sensitivity.
Transport Timelines	**Liquid:** Examine in **30 mins** (trophozoites die fast). **Formed:** Examine in **1 hour** (cysts are stable).
Stool Consistency	**Watery:** Trophozoites. **Formed:** Cysts. **Mucoid:** Intestinal irritation. **Bloody:** Invasive parasites.

🧪 IV. Comprehensive Stool Preservatives

Preservative Type	Key Characteristics & Diagnostic Uses	Limitations
10% Formalin	All-purpose fixative. Preserves eggs, cysts, and larvae well. Gold standard for **concentration** techniques (FECT).	Poor for permanent stained smears.

Preservative Type	Key Characteristics & Diagnostic Uses	Limitations
PVA (Polyvinyl Alcohol)	Best for preserving **trophozoites** for permanent stained smears (Trichrome). Highly adhesive to slides.	Toxic mercuric chloride. Poor for concentration.
SAF	Environmentally safer. Good for BOTH concentration and permanent stained smears (using iron hematoxylin).	Requires albumin to stick to slides.
MIF	Fixes and stains simultaneously. Good for field surveys. Cysts and trophozoites stain immediately.	Iodine interferes with other permanent stains.
Schaudinn's Fluid	Gold standard fixative for **fresh stool** intended for permanent stained smears. Preserves fine cytological details.	Contains toxic mercuric chloride. Requires fresh specimen.

👉 **TWO-VIAL SYSTEM: Standard protocol uses one vial of 10% Formalin (for concentration) and one vial of PVA (for permanent stains).**

🔬 V. Diagnostic Parasitology Tests & Techniques

Test / Technique	Primary Purpose & Mechanism	Target Parasites / Key Notes
1. Direct Fecal Smear (DFS)	Using 0.85% NaCl (saline) or Lugol's Iodine. Saline evaluates **motility**. Iodine stains cyst nuclei.	Detects motile *Giardia lamblia* or *Entamoeba histolytica*.
2. Kato Thick Smear	Cellophane soaked in glycerine (clearing agent) and malachite green (reduces glare). Uses 50mg feces.	Good for **thick-shelled eggs** (*Ascaris lumbricoides*).
3. Kato-Katz Technique	Quantitative modification of Kato Thick using a standardized template. Results in **Eggs Per Gram (EPG)**.	WHO standard for Soil-Transmitted Helminths & *Schistosoma mansoni*.
4. FECT (Concentration)	Sedimentation technique based on specific gravity. Centrifugation drives heavy eggs/cysts to bottom.	Standard for all eggs, cysts, larvae. Recovers **operculated eggs**.
5. Zinc Sulfate Flotation	Uses $ZnSO_4$ with Specific Gravity of 1.18. Parasites float to top meniscus.	Protozoan cysts/light eggs. Misses operculated/heavy eggs.
6. Sheather's Sugar	Boiled sugar solution with a high specific gravity (1.27).	Recommended for **Cryptosporidium parvum** oocysts.
7. Baermann Technique	Relies on hydrotropism and thermotropism. Larvae migrate from stool through warm water funnel.	Recovery of **Strongyloides stercoralis** larvae.
8. Harada-Mori Culture	Capillary action draws water up filter paper; larvae hatch and migrate to water at the bottom.	Differentiates **Hookworm** vs. **Strongyloides** larvae.
9. Agar Plate Culture	Stool placed on agar. Larvae crawl carrying bacteria, leaving visible "tracks".	Highly sensitive for *Strongyloides stercoralis*.
10. Scotch Tape Swab	Clear cellulose tape pressed against perianal folds early morning before washing.	Diagnostic test of choice for **Enterobius vermicularis** and *Taenia* eggs.
11. Enterotest (String Test)	Patient swallows capsule with nylon string. Pulled out after 4 hours to sample duodenal fluid.	Detects *Giardia lamblia*, *Strongyloides stercoralis*, and biliary flukes.
12. Thick/Thin Blood Smears	**Thick:** Drops of blood dehemoglobinized. **Thin:** Fixed in methanol (species ID).	Gold standard for *Plasmodium falciparum*, *Babesia microti*.
13. Knott's Concentration	Lyses RBCs with 2% formalin and centrifuges to concentrate blood parasites.	Detection of **Microfilariae** (*Wuchereria bancrofti*).
14. Xenodiagnosis	Allowing uninfected reduviid bugs to feed on patient, examining bug feces later.	Historical diagnostic for **Chagas disease** (*Trypanosoma cruzi*).
15. NNN Medium Culture	Novy-MacNeal-Nicolle biphasic blood agar medium.	Cultivation of *Leishmania donovani* and *Trypanosoma cruzi*.
16. Boeck & Drbohlav's	Egg-yolk based solid phase with Locke's solution liquid phase.	Specialized culture medium for *Entamoeba histolytica*.

👉 **FLOTATION VS SEDIMENTATION: Flotation = cleaner prep but misses heavy eggs. Sedimentation = recovers ALL eggs but has more debris.**

🐍 VI. Nematodes (Roundworms) - Individual Species Outline

Parasite & Name	Stages & M.O.T.	Hosts & Cycle	Distinguishing Morphological Characteristics	Pathogenesis & Clinical Info
Ascaris lumbricoides (Giant Intest. Roundworm)	**IS:** Embryonated Egg **DS:** Unembryonated Egg **MOT:** Fecal-oral	**DH:** Human **IH:** None **CYCLE:** Direct	• Largest nematode, 3 lips. Secretes **pepsin inhibitor**. • **Fertilized Egg:** Thick shell, **mammillated (bumpy) coat**. • **Unfertilized Egg:** Elongated, thinner shell. May be decorticated.	• Lung migration causes **Loeffler's syndrome** (eosinophilic pneumonia). • Intestinal obstruction or biliary ascariasis. **Charcot-Leyden** crystals.
Trichuris trichiura (Whipworm)	**IS:** Embryonated Egg **DS:** Unembryonated Egg **MOT:** Fecal-oral	**DH:** Human **IH:** None **CYCLE:** Direct	• Slender anterior (**whip**) embeds in cecum, thick posterior. • **Egg:** **Barrel/football shape** with clear mucoid **bipolar plugs**.	• NO lung migration. Capillary blood feeders in the colon. • Chronic dysentery, severe anemia, and **Rectal Prolapse**.
Necator americanus (New World Hookworm)	**IS:** Filariform (L3) **DS:** Thin-shelled Egg **MOT:** Skin Penetration	**DH:** Human **IH:** None **CYCLE:** Direct	• S-shaped body. Buccal cavity has **cutting plates**. • **Egg:** Highly transparent thin shell, typically in 4-8 cell stage.	• Blood suckers causing **Microcytic Hypochromic Anemia** (Iron Deficiency). • Skin penetration causes **Ground Itch**.

Parasite & Name	Stages & M.O.T.	Hosts & Cycle	Distinguishing Morphological Characteristics	Pathogenesis & Clinical Info
Ancylostoma duodenale (Old World Hookworm)	**IS:** Filariform (L3) **DS:** Thin-shelled Egg **MOT:** Skin Pen. / Oral	**DH:** Human **IH:** None **CYCLE:** Direct	• C-shaped body. Buccal cavity has **ventral teeth**. • **Egg:** Morphologically identical to *N. americanus*.	• More pathogenic blood sucker than *N. americanus*. • Ingestion of L3 causes **Wakana disease** (pharyngeal symptoms).
Ancylostoma ceylanicum (Zoonotic Hookworm)	**IS:** Filariform (L3) **DS:** Thin-shelled Egg **MOT:** Skin Penetration	**DH:** Cats/Dogs, Humans **IH:** None **CYCLE:** Direct	• Similar to *A. duodenale* but smaller. Distinct teeth arrangement.	• Zoonotic hookworm capable of producing patent (egg-laying) intestinal infections in humans.
Ancylostoma braziliense (Cat/Dog Hookworm)	**IS:** Filariform (L3) **DS:** Clinical appearance **MOT:** Skin Penetration	**DH:** Cats/Dogs **ACCIDENTAL:** Human	• Larvae penetrate human skin but cannot complete life cycle in humans (dead-end).	• Primary cause of **Cutaneous Larva Migrans (Creeping Eruption)**. Serpiginous itchy tracts.
Strongyloides stercoralis (Threadworm)	**IS:** Filariform (L3) **DS:** Rhabditiform (L1) **MOT:** Skin / Autoinfect	**DH:** Human **IH:** None **CYCLE:** Direct/Complex	• Females reproduce via **parthenogenesis**. Free-living soil cycle. • **L1 Larva:** **Short buccal cavity** & prominent genital primordium. • **Egg:** "Chinese lantern" shape, hatches in mucosa (rare in stool).	• Capable of massive internal **Autoinfection**. • **Hyperinfection syndrome** fatal in immunocompromised.
Enterobius vermicularis (Pinworm / Seatworm)	**IS:** Embryonated Egg **DS:** Egg on perianal fold **MOT:** Oral / Autoinfect	**DH:** Human **IH:** None **CYCLE:** Direct	• Adult has pointed tail. Both sexes have prominent **cephalic alae**. • **Egg:** Distinctly **"D-shaped"**. Thick shell. Infective in hours.	• Female migrates out of anus at night; intense **Pruritus ani**. • Highly contagious. Diagnosed via **Scotch tape swab**.
Capillaria philippinensis (Pudoc Worm)	**IS:** Infective Larva **DS:** Peanut-shaped Egg **MOT:** Oral (raw fish)	**DH:** Human, Birds **IH:** Freshwater Fish **CYCLE:** Indirect	• **Egg:** Resembles *Trichuris trichiura* but **peanut-shaped** with flattened bipolar plugs and striated shell.	• Causes severe **protein-losing enteropathy**. • Autoinfection leads to fatal cachexia/malabsorption.
Trichinella spiralis (Muscle Worm)	**IS:** Encysted Larva **DS:** Encysted Larva **MOT:** Oral (raw pork/bear)	**DH & IH:** Same host (Human, Pig, Rat) **CYCLE:** Direct (Zoonotic)	• Viviparous. Host acts as BOTH definitive and intermediate host. • Larvae encyst in **striated nurse cells** (muscle).	• Trichinosis. Enteral phase (GI distress) followed by Parenteral phase (myalgia, **periorbital edema**, high eosinophilia).
Wuchereria bancrofti (Bancroft's Filaria)	**IS:** L3 Larva **DS:** Microfilaria in blood **MOT:** Mosquito bite	**DH:** Human **IH/VECTOR:** Culex, Aedes, Anopheles	• **Microfilaria:** Sheathed. Nuclei do **NOT extend to the tip** of the tail. Exhibits **nocturnal periodicity**.	• Adults live in lymphatics. Causes **Elephantiasis** (Bancroftian filariasis) affecting lower extremities.
Brugia malayi (Malayan Filaria)	**IS:** L3 Larva **DS:** Microfilaria in blood **MOT:** Mosquito bite	**DH:** Human **IH/VECTOR:** Mansonia, Aedes	• **Microfilaria:** Sheathed. Tail has 2 distinct **terminal nuclei** separated by a space. Kinked tail.	• Malayan filariasis. Edema restricted to areas **below the knee/elbow**.
Loa loa (African Eye Worm)	**IS:** L3 Larva **DS:** Microfilaria in blood **MOT:** Fly bite	**DH:** Human **IH/VECTOR:** Chrysops dimidiata (Deer fly)	• **Microfilaria:** Sheathed. Nuclei extend **continuously to the tip**. Diurnal periodicity.	• Adults migrate through subconjunctival tissue (visible in eye). Causes transient **Calabar swellings**.
Onchocerca volvulus (Blinding Filaria)	**IS:** L3 Larva **DS:** Microfilaria in tissue **MOT:** Fly bite	**DH:** Human **IH/VECTOR:** Simulium damnosum (Black fly)	• **Microfilaria:** Unsheathed. Nuclei do not extend to tip. Diagnosed via **skin snip** (not blood).	• River Blindness (Onchocerciasis). Causes fibrous skin nodules and severe **ocular lesions**.
Mansonella perstans (Serous Cavity Filaria)	**IS:** L3 Larva **DS:** Microfilaria in blood **MOT:** Midge bite	**DH:** Human **IH/VECTOR:** Culicoides species (Midge)	• **Microfilaria:** Unsheathed. Nuclei extend to the tip of the tail. Non-periodic.	• Often asymptomatic. Can cause mild allergic symptoms and pruritus.
Dracunculus medinensis (Guinea Fire Worm)	**IS:** L3 Larva in copepod **DS:** Adult / Larva in ulcer **MOT:** Oral (drinking water)	**DH:** Human **IH:** Cyclops (Copepod) **CYCLE:** Indirect	• Longest nematode of humans. Female protrudes from ulcer to release larvae into water.	• Causes painful burning skin ulcers. Historically extracted by rolling on a stick.
Toxocara canis (Dog Roundworm)	**IS:** Embryonated Egg **DS:** Larva in tissue **MOT:** Oral (dog feces)	**DH:** Dogs **ACCIDENTAL:** Human	• Humans are **accidental dead-end hosts**. Larvae cannot mature into adults in humans.	• **Visceral Larva Migrans (VLM)**. Wandering larvae cause severe inflammation, granulomas in liver/eye.
Toxocara cati (Cat Roundworm)	**IS:** Embryonated Egg **DS:** Larva in tissue **MOT:** Oral (cat feces)	**DH:** Cats **ACCIDENTAL:** Human	• Adults in cats feature broad arrow-shaped cervical alae.	• Similar to *T. canis*, causes Visceral Larva Migrans (VLM) in humans.

VII. Cestodes (Tapeworms) - Individual Species Outline

Parasite & Name	Stages & M.O.T.	Hosts & Cycle	Distinguishing Morphological Characteristics	Pathogenesis & Clinical Info
Taenia solium (Pork Tapeworm)	**IS:** Cysticercus / Egg **DS:** Egg / Gravid Proglot. **MOT:** Ingestion	**DH:** Human **IH:** Pig **ACCIDENTAL:** Human	• **Scolex:** **ARMED.** Rostellum with 2 rows of hooks + 4 suckers. • **Proglottid:** **< 13 lateral uterine branches**. • **Egg:** Radially striated embryophore, 6-hooked oncosphere.	• Ingesting pork larva = Taeniasis (intestinal). • Ingesting **EGGS** = **Neurocysticercosis** (brain cysts, seizures).
Taenia saginata (Beef Tapeworm)	**IS:** Cysticercus bovis **DS:** Egg / Gravid Proglot. **MOT:** Ingestion	**DH:** Human **IH:** Cattle **CYCLE:** Indirect	• **Scolex:** **UNARMED.** 4 strong suckers, no hooks. • **Proglottid:** **> 15 lateral uterine branches**. Active motility. • **Egg:** Indistinguishable from *T. solium*.	• Generally asymptomatic. Does **NOT** cause human cysticercosis. • Crawling proglottids in undergarments.
Taenia asiatica (Asian Tapeworm)	**IS:** Cysticercus viscerotropica **DS:** Egg / Gravid Proglot. **MOT:** Oral (Pig Liver)	**DH:** Human **IH:** Pig **CYCLE:** Indirect	• Shares morphology of *T. saginata* adults but life cycle uses pigs like *T. solium*. • **Scolex:** Unarmed but has prominent suckers.	• Intestinal taeniasis acquired specifically from eating raw/ undercooked **pig liver**.
Diphyllobothrium latum (Broad Fish Tapeworm)	**IS:** Plerocercoid larva **DS:** Operculated Egg **MOT:** Ingestion (raw fish)	**DH:** Human **IH 1:** Copepod **IH 2:** Freshwater Fish	• **Scolex:** **Spatulate** with 2 slit-like grooves (**bothria**). • **Proglottid:** **Rosette-shaped uterus**. Wider than long. • **Egg:** **Operculum** and a small **abopercular knob**.	• Extraordinary affinity for **Vitamin B12**. • Causes **Megaloblastic Anemia** and neurological issues.
Spirometra mansonoides (Sparganosis Tapeworm)	**IS:** Plerocercoid larva **DS:** Larva in tissue **MOT:** Oral (raw frog/snake)	**DH:** Dogs/Cats **ACCIDENTAL:** Human	• Similar life cycle to *D. latum*. Humans ingest the plerocercoid (sparganum) which migrates.	• Causes **Sparganosis**. Migrating subcutaneous nodules. Can infect eyes (Ocular sparganosis).
Echinococcus granulosus (Hydatid Tapeworm)	**IS:** Egg **DS:** Hydatid Cyst (Imaging) **MOT:** Ingestion (dog feces)	**DH:** Dogs / Canines **IH:** Sheep / Herbivores **ACCIDENTAL:** Human	• **Adult:** Strobila has only 3 proglottids. In canines only. • **Larva:** Massive Unilocular Hydatid Cyst containing **Hydatid Sand** (free protoscolices).	• Cysts primarily in liver/lungs. • Accidental rupture causes fatal **Anaphylactic Shock**.
Echinococcus multilocularis (Alveolar Hydatid Worm)	**IS:** Egg **DS:** Alveolar Cyst **MOT:** Oral (fox/dog feces)	**DH:** Foxes, Dogs **IH:** Rodents **ACCIDENTAL:** Human	• Larva forms an invasive, multi-chambered (**alveolar**) cyst without a distinct limiting membrane.	• Alveolar echinococcosis. Cyst grows aggressively in the liver, mimicking a malignant carcinoma. Often fatal.
Hymenolepis nana (Dwarf Tapeworm)	**IS:** Egg / Cysticercoid **DS:** Egg in stool **MOT:** Oral / Autoinfect	**DH:** Human, Rodents **IH:** Flea/Beetle (Optional) **CYCLE:** Direct/Indirect	• The ONLY human tapeworm requiring **NO intermediate host**. • **Egg:** Contains **polar filaments** between inner and outer membranes.	• Most common tapeworm worldwide. Causes enteritis. • Highly capable of internal **autoinfection**.
Hymenolepis diminuta (Rat Tapeworm)	**IS:** Cysticercoid **DS:** Egg in stool **MOT:** Oral (ingest insect)	**DH:** Rats (Human=Accidental) **IH:** Flea / Grain beetle **CYCLE:** Indirect	• **Egg:** Larger than *H. nana*, **LACKS polar filaments**. • **Scolex:** Unarmed (unlike *H. nana* which is armed).	• Mild intestinal distress. Typically seen in children who accidentally ingest infected grain insects.
Dipylidium caninum (Dog/Flea Tapeworm)	**IS:** Cysticercoid **DS:** Egg packets / Proglot. **MOT:** Oral (ingest flea)	**DH:** Dogs, Cats, Humans **IH:** Dog/Cat Flea **CYCLE:** Indirect	• **Proglottid:** Barrel-shaped, resembles a **pumpkin seed**. Has bilateral genital pores. • **Egg:** Passed as enclosed **egg packets** (8-15 eggs).	• Dipylidiasis. Often asymptomatic. Common in children kissing pets or accidentally swallowing fleas.

VIII. Trematodes (Flukes) - Individual Species Outline

Parasite & Name	Stages & M.O.T.	Hosts & Cycle	Distinguishing Morphological Characteristics	Pathogenesis & Clinical Info
Fasciola hepatica (Sheep Liver Fluke)	**IS:** Metacercaria **DS:** Operculated Egg **MOT:** Oral (watercress)	**DH:** Sheep, Human **IH 1:** Snail (Lymnaea) **IH 2:** Aquatic vegetation	• **Adult:** Broad, leaf-like. Anterior **cephalic cone** ("shoulders"). • **Egg:** Exceptionally large, unembryonated, distinct operculum.	• Adults in biliary passages cause hepatic colic and "liver rot". • **Halzoun syndrome:** Pharyngeal fascioliasis from raw liver.
Fasciola gigantica (Giant Liver Fluke)	**IS:** Metacercaria **DS:** Operculated Egg **MOT:** Oral (water plants)	**DH:** Cattle, Human **IH 1:** Snail **IH 2:** Aquatic vegetation	• Morphologically similar to *F. hepatica* but significantly larger and more elongated. Shorter cephalic cone.	• Causes Tropical Fascioliasis. Pathology identical to *F. hepatica*.
Schistosoma mansoni (Manson's Blood Fluke)	**IS:** Cercaria **DS:** Spined Egg in Stool **MOT:** Skin penetration	**DH:** Human **IH:** Biomphalaria snail **CYCLE:** Indirect	• **Adult:** Dioecious. Resides in inferior mesenteric venules. • **Egg:** Non-operculated. Prominent **large lateral spine**.	• Trapped eggs induce severe **Granuloma** formation and **pipe-stem fibrosis** of the liver. Intestinal schistosomiasis.
Schistosoma japonicum (Oriental Blood Fluke)	**IS:** Cercaria **DS:** Spined Egg in Stool **MOT:** Skin penetration	**DH:** Human **IH:** Oncomelania snail **CYCLE:** Indirect	• **Adult:** Resides in superior mesenteric venules. • **Egg:** Non-operculated. Small, inconspicuous **lateral knob**.	• Highest egg output. Most severe systemic granulomas. High association with **Katayama fever**.

Parasite & Name	Stages & M.O.T.	Hosts & Cycle	Distinguishing Morphological Characteristics	Pathogenesis & Clinical Info
Schistosoma haematobium (Vesical Blood Fluke)	**IS:** Cercaria **DS:** Spined Egg in Urine **MOT:** Skin penetration	**DH:** Human **IH:** Bulinus snail **CYCLE:** Indirect	• **Adult:** Resides in vesicular/pelvic venous plexus. • **Egg:** Non-operculated. Prominent **terminal spine**.	• Urinary schistosomiasis (hematuria). Highly associated with **Squamous Cell Carcinoma** of the bladder.
Paragonimus westermani (Oriental Lung Fluke)	**IS:** Metacercaria **DS:** Operculated Egg **MOT:** Oral (raw crabs)	**DH:** Human, Felines **IH 1:** Snail (Antemelania) **IH 2:** Crabs	• **Adult:** Reddish-brown, resembles a **coffee bean**. • **Egg:** Operculated, features a **thickened abopercular end**. Found in sputum or stool.	• Adults encapsulate in lung pairs. Symptoms closely mimic **Pulmonary Tuberculosis** (bloody/rusty sputum).
Clonorchis sinensis (Chinese Liver Fluke)	**IS:** Metacercaria **DS:** Operculated Egg **MOT:** Oral (raw fish)	**DH:** Human **IH 1:** Snail **IH 2:** Freshwater Fish	• **Egg:** Very small, pitcher-shaped. Distinct prominent opercular **shoulders** and a small abopercular knob.	• Adults live in bile ducts. Chronic infection associated with **Cholangiocarcinoma** (bile duct cancer).
Opisthorchis felineus (Cat Liver Fluke)	**IS:** Metacercaria **DS:** Operculated Egg **MOT:** Oral (raw fish)	**DH:** Cats, Dogs, Humans **IH 1:** Snail **IH 2:** Freshwater Fish	• Very similar to *C. sinensis*. Narrower body. Egg lacks prominent opercular shoulders.	• Opisthorchiasis. Causes cholangitis and biliary obstruction similar to *Clonorchis*.
Fasciolopsis buski (Giant Intestinal Fluke)	**IS:** Metacercaria **DS:** Operculated Egg **MOT:** Oral (water chestnut)	**DH:** Human, Pigs **IH 1:** Snail **IH 2:** Aquatic vegetation	• **Adult:** Largest intestinal fluke. **Lacks the cephalic cone** seen in *F. hepatica*. • **Egg:** Morphologically identical to *Fasciola hepatica*.	• Resides in the small intestine. Causes severe mucosal ulceration, diarrhea, and generalized edema.
Heterophyes heterophyes	**IS:** Metacercaria **DS:** Operculated Egg **MOT:** Oral (raw fish)	**DH:** Human, Cats, Dogs **IH 1:** Snail **IH 2:** Freshwater Fish	• **Adult:** Extremely small intestinal flukes. Has a 3rd sucker (**gonotyl**). • **Egg:** Very small, resembles *Clonorchis* but less distinct shoulders.	• Mild intestinal symptoms. Rarely, eggs enter circulation and cause cardiac or neural granulomas.
Metagonimus yokogawai	**IS:** Metacercaria **DS:** Operculated Egg **MOT:** Oral (raw fish)	**DH:** Human, Cats, Dogs **IH 1:** Snail **IH 2:** Freshwater Fish	• **Adult:** Smallest fluke infecting humans. Ventral sucker is displaced to the right side of the body.	• Metagonimiasis. Clinically identical to *Heterophyes* (diarrhea, potential erratic egg migration).

🦠 IX. Protozoa (Amoebae & Ciliates) - Complete Species Outline

Parasite & Name	Stages & M.O.T.	Hosts & Cycle	Distinguishing Morphological Characteristics	Pathogenesis & Clinical Info
Entamoeba histolytica (Pathogenic Amoeba)	**IS:** Mature Cyst **DS:** Cyst / Trophozoite **MOT:** Fecal-oral	**DH:** Human **IH:** None **CYCLE:** Direct	• **Trophozoite:** Progressive directional motility. May contain **ingested RBCs** (diagnostic). Central karyosome. • **Cyst:** Contains 1 to 4 nuclei. Has **cigar-shaped** (rounded ends) chromatoidal bars.	• Causes Amoebic Dysentery. Destroys intestinal mucosa causing **flask-shaped ulcers**. • Extraintestinal: Amoebic Liver Abscess yielding **"anchovy sauce"** aspirate.
Entamoeba hartmanni (Small Race E. histolytica)	**IS:** Mature Cyst **DS:** Cyst / Trophozoite **MOT:** Fecal-oral	**DH:** Human **IH:** None **CYCLE:** Direct	• Morphologically identical to *E. histolytica* but **significantly smaller** (<10 µm cyst). Never ingests RBCs.	• **Non-pathogenic** commensal. Important to measure via ocular micrometer to avoid false positive for *E. histolytica*.
Entamoeba coli (Commensal Amoeba)	**IS:** Mature Cyst **DS:** Cyst / Trophozoite **MOT:** Fecal-oral	**DH:** Human **IH:** None **CYCLE:** Direct	• **Trophozoite:** Sluggish, non-directional motility. Contains ingested bacteria (never RBCs). Eccentric karyosome. • **Cyst:** Contains **1 to 8 nuclei** (mature). Has **splinter-like** (pointed ends) chromatoidal bars.	• **Non-pathogenic** commensal. Indicates ingestion of fecally contaminated material.
Entamoeba polecki (Pig/Monkey Amoeba)	**IS:** Mature Cyst **DS:** Cyst / Trophozoite **MOT:** Fecal-oral (pig feces)	**DH:** Pigs, Monkeys **ACCIDENTAL:** Human	• **Cyst:** Consistently possesses only **one nucleus**. Pleomorphic chromatoidal bars with angular/pointed ends.	• Generally considered a **non-pathogenic** commensal in humans, but rare cases of mild diarrhea reported.
Entamoeba gingivalis (Oral Amoeba)	**IS:** Trophozoite **DS:** Trophozoite in saliva **MOT:** Kissing / Utensils	**DH:** Human **IH:** None **CYCLE:** Direct	• **Trophozoite:** Found in the mouth/gum pockets. The only amoeba that actively **ingests WBCs**. • **NO CYST STAGE** exists.	• **Non-pathogenic**. Often found abundantly in patients with periodontal disease or poor oral hygiene.
Endolimax nana (Dwarf Amoeba)	**IS:** Mature Cyst **DS:** Cyst / Trophozoite **MOT:** Fecal-oral	**DH:** Human **IH:** None **CYCLE:** Direct	• **Trophozoite:** Very small. Nucleus has a large, irregular karyosome with no peripheral chromatin. • **Cyst:** Oval shaped. Contains 4 nuclei with large karyosomes (gives a **"cross-eyed" cyst** appearance).	• **Non-pathogenic** commensal. Smallest of the intestinal amoebae.

Parasite & Name	Stages & M.O.T.	Hosts & Cycle	Distinguishing Morphological Characteristics	Pathogenesis & Clinical Info
Iodamoeba bütschlii (Iodine Cyst)	**IS:** Mature Cyst **DS:** Cyst / Trophozoite **MOT:** Fecal-oral	**DH:** Human, Pigs **IH:** None **CYCLE:** Direct	• **Trophozoite:** Nucleus has a large karyosome surrounded by achromatic granules (**basket nucleus**). • **Cyst:** Contains only 1 nucleus. Characterized by a massive, compact **glycogen vacuole** that stains deep mahogany brown with Iodine.	• **Non-pathogenic** commensal. The glycogen vacuole is the key diagnostic feature.
Naegleria fowleri (Brain-Eating Amoeba)	**IS:** Trophozoite **DS:** Trophozoite in CSF **MOT:** Nasal Inhalation	**DH:** Human (Accidental) **ENV:** Warm freshwater	• Free-living amoeba. Can transform into a **flagellated form** in warm water. • **Trophozoite:** Rapid, eruptive (lobopodial) movement. NO cysts found in human tissue.	• Enters via olfactory nerve. Causes **Primary Amoebic Meningoencephalitis (PAM)**. Rapidly fatal within days.
Acanthamoeba castellanii (Free-living Amoeba)	**IS:** Trophozoite / Cyst **DS:** Troph / Cyst in tissue **MOT:** Broken skin/eyes	**DH:** Human (Accidental) **ENV:** Soil/Water	• **Trophozoite:** Possesses distinct spine-like pseudopods (**acanthopodia**). • **Cyst:** Double-walled with a wrinkled outer layer (exocyst) and polygonal inner layer (endocyst).	• Causes **Granulomatous Amoebic Encephalitis (GAE)** in immunocompromised. • Causes severe Amoebic Keratitis (associated heavily with **contact lens** use).
Balamuthia mandrillaris (Free-living Amoeba)	**IS:** Trophozoite / Cyst **DS:** Troph / Cyst in tissue **MOT:** Inhalation / Soil contact	**DH:** Human, Mandrills **ENV:** Soil/Dust	• Morphologically similar to *Acanthamoeba* but cysts have three wall layers. Broad, flat pseudopodia.	• Causes **Granulomatous Amoebic Encephalitis (GAE)** in both healthy and immunocompromised hosts. Often features skin lesions prior to CNS symptoms.
Balantidium coli (Intestinal Ciliate)	**IS:** Mature Cyst **DS:** Cyst / Trophozoite **MOT:** Fecal-oral	**DH:** Pigs (Reservoir) **ACCIDENTAL:** Human **CYCLE:** Direct	• **Largest** parasitic protozoan. Covered entirely in cilia. Exhibits rotary/boring motility. • Contains two nuclei: a large **kidney-shaped macronucleus** and a tiny round micronucleus.	• Balantidiasis. Zoonotic from pigs. The ONLY ciliate pathogenic to humans. Invades mucosa causing ulcers and **dysentery** similar to *E. histolytica*.

🦠 X. Protozoa (Flagellates) - Complete Species Outline

Parasite & Name	Stages & M.O.T.	Hosts & Cycle	Distinguishing Morphological Characteristics	Pathogenesis & Clinical Info
Giardia lamblia (Traveler's Diarrhea)	**IS:** Mature Cyst **DS:** Cyst / Trophozoite **MOT:** Fecal-oral / Water	**DH:** Human **RESERVOIR:** Beavers **CYCLE:** Direct	• **Trophozoite:** Pear-shaped, 2 nuclei, 4 pairs of flagella. Looks like an **"old man with glasses"**. Has **falling leaf motility**. Sucking disc. • **Cyst:** Oval, contains 4 nuclei and retracted flagella/axonemes.	• Giardiasis. Attaches to duodenal mucosa blocking fat absorption. • Causes **Steatorrhea** (fatty, foul-smelling stool) and "Gay Bowel Syndrome".
Dientamoeba fragilis (Amoeba-like Flagellate)	**IS:** Trophozoite (inside helminth egg) **DS:** Trophozoite in stool **MOT:** Fecal-oral	**DH:** Human **IH:** None **CYCLE:** Direct	• Classified as a flagellate structurally, but moves like an amoeba and LACKS flagella. • **Trophozoite:** Binucleated (majority). Nuclei have fragmented karyosomes (**rosette shape**). • **NO CYST STAGE**.	• Pathogenic in heavy loads. Causes intermittent diarrhea. • Highly associated with co-infection and transmission via the eggs of **Enterobius vermicularis** (pinworm).
Chilomastix mesnili (Intestinal Commensal)	**IS:** Cyst **DS:** Cyst / Trophozoite **MOT:** Fecal-oral	**DH:** Human **IH:** None **CYCLE:** Direct	• **Trophozoite:** Pear-shaped, distinctive spiral groove. Exhibits stiff, rotary motility. • **Cyst:** Characteristic **lemon shape** with a clear anterior hyaline knob (nipple). Contains a cytostome resembling a **safety pin** (shepherd's crook).	• **Non-pathogenic** commensal. Important to distinguish from Giardia.
Pentatrichomonas hominis (Intestinal Trichomonad)	**IS:** Trophozoite **DS:** Trophozoite in stool **MOT:** Fecal-oral	**DH:** Human **IH:** None **CYCLE:** Direct	• **Trophozoite:** Undulating membrane extends the **entire length** of the body with a trailing free flagellum. • **NO CYST STAGE**.	• **Non-pathogenic** commensal of the large intestine. Indicator of fecal contamination.
Trichomonas vaginalis (Urogenital Flagellate)	**IS:** Trophozoite **DS:** Trophozoite **MOT:** Sexual contact	**DH:** Human **IH:** None **CYCLE:** Direct	• **Trophozoite:** Pear-shaped, undulating membrane extends **HALF** the body length. • Exhibits rapid, **jerky tumbling motility** in fresh wet mounts. • **NO CYST STAGE** exists.	• Trichomoniasis (STD). Causes vaginitis with frothy, yellowish-green discharge. Males often asymptomatic carriers. • Clinical hallmark: Punctate hemorrhages on cervix (**"Strawberry cervix"**).
Trichomonas tenax (Oral Trichomonad)	**IS:** Trophozoite **DS:** Trophozoite in saliva **MOT:** Kissing / Utensils	**DH:** Human **IH:** None **CYCLE:** Direct	• **Trophozoite:** Smallest of the 3 trichomonads. Undulating membrane extends **two-thirds** of the body length. • **NO CYST STAGE**. Found in tartar and gum crevices.	• **Non-pathogenic** oral commensal. Associated with poor oral hygiene and periodontal disease.

Parasite & Name	Stages & M.O.T.	Hosts & Cycle	Distinguishing Morphological Characteristics	Pathogenesis & Clinical Info
Trypanosoma brucei gambiense (West African Sleeping Sickness)	**IS:** Metacyclic Trypomastigote **DS:** Trypomastigote in blood **MOT:** Fly bite	**DH:** Human **IH/VECTOR:** Glossina palpalis (Tsetse) **CYCLE:** Indirect	• **Trypomastigote:** Found in blood/CSF. Has a central nucleus, posterior kinetoplast, and an undulating membrane extending the FULL body length.	• Causes **Chronic** African Sleeping Sickness. Invades CNS causing prolonged somnolence/coma. • Enlarged posterior cervical lymph nodes (**Winterbottom's sign**).
Trypanosoma brucei rhodesiense (East African Sleeping Sickness)	**IS:** Metacyclic Trypomastigote **DS:** Trypomastigote in blood **MOT:** Fly bite	**DH:** Human, Wild Game **IH/VECTOR:** Glossina morsitans (Tsetse)	• Morphologically identical to *T.b. gambiense* in human blood smears.	• Causes **Acute** African Sleeping Sickness. Rapid progression to severe CNS involvement and death. Zoonotic.
Trypanosoma cruzi (American Trypanosomiasis)	**IS:** Metacyclic Trypomastigote **DS:** Trypomastigote, Amastigote **MOT:** Bug feces in bite wound	**DH:** Human **IH/VECTOR:** Triatoma infestans (Reduviid) **CYCLE:** Indirect	• **Trypomastigote:** Assumes a characteristic **"C" or "U" shape** in blood smears. Large kinetoplast. • **Amastigote:** Intracellular stage found heavily in cardiac/smooth muscle. Lacks external flagellum.	• Causes **Chagas Disease**. Transmitted by the "Kissing Bug". • Unilateral periorbital edema (**Romaña's sign**). Chronic phase causes megacolon, megaesophagus, and fatal cardiomyopathy.
Leishmania donovani (Visceral Leishmaniasis)	**IS:** Promastigote **DS:** Amastigote in macrophages **MOT:** Fly bite	**DH:** Human, Dogs **IH/VECTOR:** Phlebotomus (Sandfly) **CYCLE:** Indirect	• **Promastigote:** Elongated, flagellated form found in the sandfly and lab cultures (NNN medium). • **Amastigote:** Oval, non-flagellated form packed inside RES macrophages. Known clinically as **Leishman-Donovan (LD) bodies**.	• Causes **Kala-azar** (Visceral Leishmaniasis). Deadly if untreated. • Massive hepatosplenomegaly, pancytopenia, spiking fevers, darkening of the facial skin.
Leishmania tropica (Old World Cutaneous)	**IS:** Promastigote **DS:** Amastigote in macrophages **MOT:** Fly bite	**DH:** Human **IH/VECTOR:** Phlebotomus (Sandfly) **CYCLE:** Indirect	• Morphologically identical to *L. donovani* but localized strictly to cutaneous macrophages at the bite site.	• Causes **Oriental Sore** (Delhi boil, Aleppo button). Localized dry, crusted skin ulcers. Self-healing with immunity.
Leishmania braziliensis (Mucocutaneous Leishmaniasis)	**IS:** Promastigote **DS:** Amastigote in tissue **MOT:** Fly bite	**DH:** Human, Forest rodents **IH/VECTOR:** Lutzomyia (Sandfly) **CYCLE:** Indirect	• Morphologically identical to other Leishmania species. Diagnosis requires tissue biopsy from mucosal edges.	• Causes **Espundia**. Severe, disfiguring erosion of the nasal septum, palate, and oral mucosa. Does not self-heal.
Leishmania mexicana (New World Cutaneous)	**IS:** Promastigote **DS:** Amastigote in macrophages **MOT:** Fly bite	**DH:** Human, Rodents **IH/VECTOR:** Lutzomyia (Sandfly) **CYCLE:** Indirect	• Morphologically identical to other Leishmania species. Amastigotes localized to skin.	• Causes **Chiclero's Ulcer** (Bay sore). Typically forms a single, painless ulcer, frequently on the ear pinna.

XI. Protozoa (Apicomplexa/Sporozoa & Microsporidia)

Parasite & Name	Stages & M.O.T.	Hosts & Cycle	Distinguishing Morphological Characteristics	Pathogenesis & Clinical Info
Plasmodium falciparum (Malignant Tertian Malaria)	**IS:** Sporozoite **DS:** Trophs / Gametocytes **MOT:** Mosquito bite	**DH:** Anopheles Mosquito **IH:** Human **CYCLE:** Indirect	• Infects RBCs of ALL ages. Normal sized RBCs. • **Trophozoite:** Appears as delicate rings; frequently features **multiple rings per RBC**, applique forms, and double chromatin dots. • **Gametocyte:** Highly distinct **crescent or banana shape**. Shows Maurer's clefts/dots.	• Most deadly form of malaria (36-48 hr fever cycle). Causes severe hemolysis and microvascular sludging. • Complications: Cerebral malaria, Algid malaria, and **Blackwater fever** (massive hemoglobinuria).
Plasmodium vivax (Benign Tertian Malaria)	**IS:** Sporozoite **DS:** All stages in blood **MOT:** Mosquito bite	**DH:** Anopheles Mosquito **IH:** Human **CYCLE:** Indirect	• Infects only RETICULOCYTES (young RBCs). Infected RBCs become **enlarged and pale**. • **Trophozoite:** Highly irregular/amoeboid shape filling the RBC. • Shows prominent eosinophilic stippling known as **Schüffner's dots**.	• Causes relapsing malaria (48 hr cycle). Parasites form dormant **hypnozoites** in the human liver, causing recurrent infection years later.
Plasmodium malariae (Quartan Malaria)	**IS:** Sporozoite **DS:** All stages in blood **MOT:** Mosquito bite	**DH:** Anopheles Mosquito **IH:** Human **CYCLE:** Indirect	• Infects only OLD RBCs. Normal to smaller RBC size. • **Trophozoite:** Extends across the RBC forming a distinct **Band form**. • **Schizont:** Arranged in a **rosette** shape (6-12 merozoites). Shows Ziemann's dots.	• Causes **72-hour** (quartan) fever cycle. • Chronic infection is strongly associated with **Nephrotic syndrome** due to immune complex deposition.
Plasmodium ovale (Ovale Tertian Malaria)	**IS:** Sporozoite **DS:** All stages in blood **MOT:** Mosquito bite	**DH:** Anopheles Mosquito **IH:** Human **CYCLE:** Indirect	• Infects young RBCs. Shows James's dots (similar to Schüffner's). • Infected RBCs become enlarged, distinctly **oval-shaped**, and feature **fimbriated (ragged) edges**.	• Clinically similar to *P. vivax* (48 hr cycle). Also forms liver **hypnozoites** leading to late relapses. Rarer, found mostly in West Africa.
Plasmodium knowlesi (Simian Malaria)	**IS:** Sporozoite **DS:** All stages in blood **MOT:** Mosquito bite	**DH:** Anopheles Mosquito **IH:** Macaque Monkeys, Humans	• Morphologically resembles *P. malariae* (band forms) but has the rapid replication cycle and severity of *P. falciparum*. • Requires PCR for definitive species identification.	• Zoonotic malaria (SE Asia). Features a rapid **24-hour** (quotidian) fever cycle. Can rapidly progress to severe, fatal disease.

Parasite & Name	Stages & M.O.T.	Hosts & Cycle	Distinguishing Morphological Characteristics	Pathogenesis & Clinical Info
Babesia microti (Babesiosis)	**IS:** Sporozoite **DS:** Trophozoite in RBC **MOT:** Tick bite / Transfusion	**DH:** Ixodes Tick **IH:** Rodents, Human **CYCLE:** Indirect	• Intraerythrocytic parasite. Resembles *Plasmodium* rings but LACKS malarial pigment (hemozoin). • Trophozoites often arranged in tetrads forming a pathognomonic **Maltese cross**. Extracellular forms may be seen.	• Causes hemolytic anemia resembling malaria. Often co-transmitted with Lyme disease by the Deer Tick. Severe in splenectomized patients.
Toxoplasma gondii (Toxoplasmosis)	**IS:** Oocyst / Tissue Cyst **DS:** Tachyzoite / Serology **MOT:** Ingestion / Congenital	**DH:** Felines (Cats) **IH:** Rodents, Pigs, Human **CYCLE:** Indirect	• **Tachyzoite:** Crescent/bow-shaped, rapidly dividing form found in acute infection. • **Bradyzoite:** Slowly dividing form enclosed in tissue cysts (muscle/brain).	• Transmitted via **cat feces** or undercooked meat. • Causes severe **Congenital Toxoplasmosis** (TORCH complex) leading to hydrocephalus, intracranial calcifications, and chorioretinitis.
Cryptosporidium parvum (Cryptosporidiosis)	**IS:** Thick-walled Oocyst **DS:** Oocyst in stool **MOT:** Fecal-oral / Water	**DH:** Human, Animals **IH:** None **CYCLE:** Direct	• **Oocyst:** Tiny (4-6 µm). Contains 4 sporozoites. Highly resistant to standard water chlorination. • Visualized distinctly using the **Modified Acid-Fast Stain** (appears red/pink against a blue background).	• Extracellular but intracellularly located in brush border. Self-limiting watery diarrhea in healthy individuals. • Causes **severe, intractable cholera-like diarrhea** in HIV/AIDS patients.
Cyclospora cayetanensis (Cyclosporiasis)	**IS:** Sporulated Oocyst **DS:** Unsporulated Oocyst **MOT:** Oral (food/water)	**DH:** Human **IH:** None **CYCLE:** Direct	• **Oocyst:** Larger than Crypto (8-10 µm). Contains 2 sporocysts, each with 2 sporozoites. • Stains **variable acid-fast** (some stain red, some remain unstained/ghost cells). Exhibits auto-fluorescence under UV light.	• Causes prolonged, relapsing watery diarrhea. Associated with imported fresh produce (raspberries, basil).
Cystoisospora belli (Isosporiasis)	**IS:** Sporulated Oocyst **DS:** Unsporulated Oocyst **MOT:** Fecal-oral	**DH:** Human **IH:** None **CYCLE:** Direct	• **Oocyst:** Large (20-30 µm), distinctive **elongated/ellipsoidal** shape. • Passed unsporulated (containing 1 sporoblast). Matures to contain 2 sporocysts. Stains Acid-Fast positive.	• Causes severe diarrhea, weight loss, and malabsorption, particularly in immunocompromised (HIV) patients. Often features high peripheral eosinophilia.
Sarcocystis bovihominis* / *suihominis	**IS:** Sarcocyst (in meat) **DS:** Sporocyst in stool **MOT:** Oral (raw beef/pork)	**DH:** Human **IH:** Cattle (bovi) / Pig (sui) **CYCLE:** Indirect	• **Sporocyst:** Oocyst wall breaks down in gut; individual sporocysts containing 4 sporozoites are passed in stool. • **Sarcocyst:** Tissue cyst found in striated muscle of intermediate host (Miescher's tubes).	• Intestinal sarcocystosis: self-limiting diarrhea/vomiting from eating undercooked beef or pork containing sarcocysts.
Enterocytozoon bieneusi ***Encephalitozoon spp.*** (Microsporidia)	**IS:** Spore **DS:** Spore in stool/tissue **MOT:** Fecal-oral / Inhalation	**DH:** Human, Animals **IH:** None **CYCLE:** Direct	• Obligate intracellular parasite (now reclassified closely with Fungi). Extremely small spores (1-2 µm). • Features a unique **coiled polar tubule** used to inject infective material into host cells. Diagnosed via Modified Trichrome (Weber's).	• *E. bieneusi* causes chronic diarrhea and wasting in HIV/AIDS. • *E. intestinalis* causes diarrhea and disseminates to the kidneys.

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🐜 XII. Medical Arthropodology (Vectors & Ectoparasites)

Arthropod & Name	Classification / M.O.T.	Pathogen Transmitted	Distinguishing Characteristics	Clinical Info & Pathogenesis
Anopheles minimus (Malaria Mosquito)	**CLASS:** Insecta **ROLE:** Biological Vector **BITE:** Nocturnal	**PARASITE:** *Plasmodium falciparum* **NEMATODE:** *Wuchereria bancrofti*	• Adults rest with their abdomen pointing upward at an **angle** to the surface. • Larvae lack a respiratory siphon and lie parallel to the water surface.	• Definitive host for Malaria (sexual reproduction occurs in the mosquito gut).
Aedes aegypti (Tiger Mosquito)	**CLASS:** Insecta **ROLE:** Biological Vector **BITE:** Diurnal (Daytime)	**VIRUSES:** Dengue, Zika, Yellow Fever **NEMATODE:** *Wuchereria bancrofti*	• Distinctive **black and white striped** markings on legs and body. • Breeds in clean, stagnant water (tires, flower pots).	• Primary vector for Dengue Hemorrhagic Fever.
Culex quinquefasciatus (Common House Mosquito)	**CLASS:** Insecta **ROLE:** Biological Vector **BITE:** Nocturnal	**VIRUSES:** Japanese Encephalitis, West Nile **NEMATODE:** *Wuchereria bancrofti*	• Brownish color. Rests parallel to the surface. • Breeds in highly polluted or dirty stagnant water.	• Major vector for urban Bancroftian filariasis and various arboviral encephalitides.
Phlebotomus papatasi (Sandfly)	**CLASS:** Insecta **ROLE:** Biological Vector **MOT:** Blood meal	**PARASITE:** *Leishmania donovani* **BACTERIA:** Bartonella bacilliformis	• Tiny, densely **hairy** flies. Wings are held erect in a V-shape when resting.	• Bite is painful. Transmits Visceral (Kala-azar) and Cutaneous Leishmaniasis.
Glossina palpalis (Tsetse Fly)	**CLASS:** Insecta **ROLE:** Biological Vector **MOT:** Blood meal	**PARASITE:** *Trypanosoma brucei gambiense*	• Wings cross like scissors over the abdomen when at rest. Prominent forward-pointing proboscis.	• Transmits **African Sleeping Sickness**. Both males and females bite and transmit the disease.
Simulium damnosum (Black Fly)	**CLASS:** Insecta **ROLE:** Biological Vector **MOT:** Blood meal	**PARASITE:** *Onchocerca volvulus*	• Small, stout flies with a humpbacked appearance. Breeds in fast-flowing streams/rivers.	• Transmits River Blindness. Bites leave a bleeding point on the skin.
Chrysops dimidiata (Deer / Mango Fly)	**CLASS:** Insecta **ROLE:** Biological Vector **MOT:** Blood meal	**PARASITE:** *Loa loa* **BACTERIA:** Francisella tularensis	• Yellow to black flies with distinct dark bands across the wings.	• Transmits the African eye worm. Painful, persistent bite.
Dermatobia hominis (Human Botfly)	**CLASS:** Insecta **ROLE:** Ectoparasite **MOT:** Larval penetration	**DISEASE:** **Myiasis** (Infection with fly larvae)	• Adult fly catches a mosquito, attaches its eggs to it. Mosquito bites human, eggs drop and hatch into larvae.	• Larvae burrow into subcutaneous tissue causing a painful, raised, furuncular lesion with a central breathing pore.
Triatoma infestans (Kissing Bug)	**CLASS:** Insecta **ROLE:** Biological Vector **MOT:** Bug feces in bite wound	**PARASITE:** *Trypanosoma cruzi*	• Large bugs with cone-shaped heads. Defecate while feeding.	• Transmits **Chagas Disease**. Often bites near the eyes/lips during sleep.
Pediculus humanus capitis (Head Louse)	**CLASS:** Insecta **ROLE:** Ectoparasite **MOT:** Direct contact	**DISEASE:** Pediculosis capitis	• Wingless, dorsoventrally flattened. Eggs are called **nits** and firmly cemented to hair shafts.	• Causes intense scalp itching. Not known to be a major disease vector like the body louse.
Pediculus humanus corporis (Body Louse)	**CLASS:** Insecta **ROLE:** Ectoparasite / Vector **MOT:** Direct contact (clothing)	**BACTERIA:** Rickettsia prowazekii, Borrelia recurrentis	• Morphologically similar to head louse but lives and lays eggs in the seams of clothing.	• Primary vector for **Epidemic Typhus** and Louse-borne Relapsing Fever.
Xenopsylla cheopis (Oriental Rat Flea)	**CLASS:** Insecta **ROLE:** Biological Vector **MOT:** Flea bite / Ingestion	**BACTERIA:** Yersinia pestis **CESTODE:** *Hymenolepis diminuta*	• Wingless, **laterally flattened**. Powerful hind legs for jumping.	• Primary vector for the **Bubonic Plague** and murine typhus.
Ixodes scapularis (Hard Tick / Deer Tick)	**CLASS:** Arachnida (8 legs) **ROLE:** Biological Vector **MOT:** Tick bite	**BACTERIA:** Borrelia burgdorferi **PARASITE:** *Babesia microti*	• Hard scutum (shield) covering the dorsal surface. Capitulum (head) is visible from above.	• Transmits **Lyme Disease** (Erythema migrans rash) and human babesiosis.
Sarcoptes scabiei (Itch Mite)	**CLASS:** Arachnida (8 legs) **ROLE:** Ectoparasite **MOT:** Direct contact	**DISEASE:** **Scabies**	• Microscopic, oval, tortoise-like body. Short stubby legs.	• Females **burrow into the stratum corneum** of the skin to lay eggs. Causes severe nocturnal itching and track-like lesions.

📚 XIII. References

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